Doiraning yuzi

Dumaloq gilam polning qancha qismini egallaydi? Dumaloq stol ustini bo'yashga qancha bo'yoq ketadi? Buni bilish uchun doiraning yuzini topamiz.

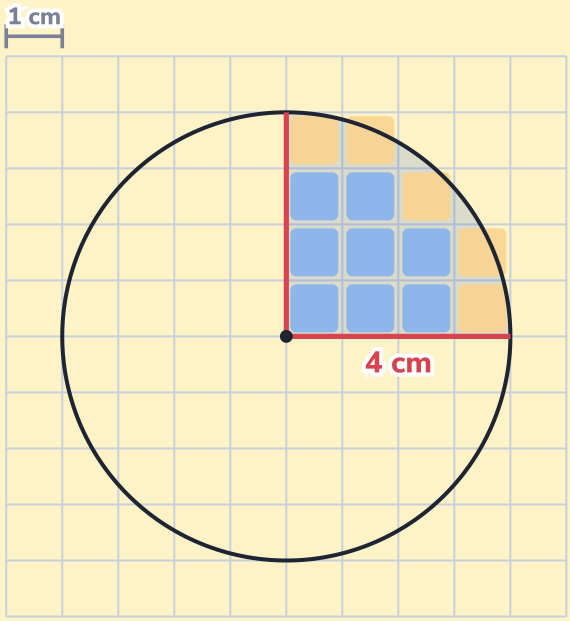
1-QISM

Mavzu

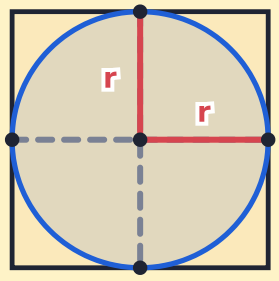
USUL · KATAKLAR BILAN CHAMALASH

Radiusi 4 cm bo'lgan doira katakli qog'ozga chizildi. Har bir katak — $1\ \text{cm}^2$.

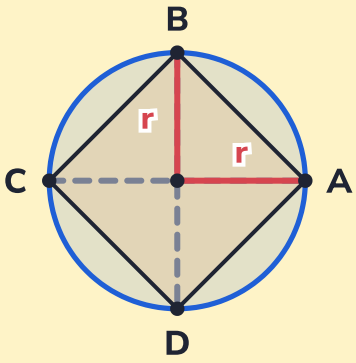
1. Doiraning choragini olamiz. Unda to'liq kataklar 8 ta (ko'k).
2. Yarmidan ko'pi chorak ichida bo'lgan kataklar 5 ta (sariq). Ularni ham butun katak deb sanaymiz. Yarmidan kami ichida bo'lganlarini tashlab ketamiz.
3. Chorak doira: $8 + 5 = 13$, taqriban $13\ \text{cm}^2$.
4. Butun doira 4 ta shunday chorakdan iborat: $4 \cdot 13 = 52\ \text{cm}^2$ (taqriban).



USUL · KVADRATLAR BILAN SOLISHTIRISH



Doira katta kvadrat ichida



Kichik kvadrat doira ichida

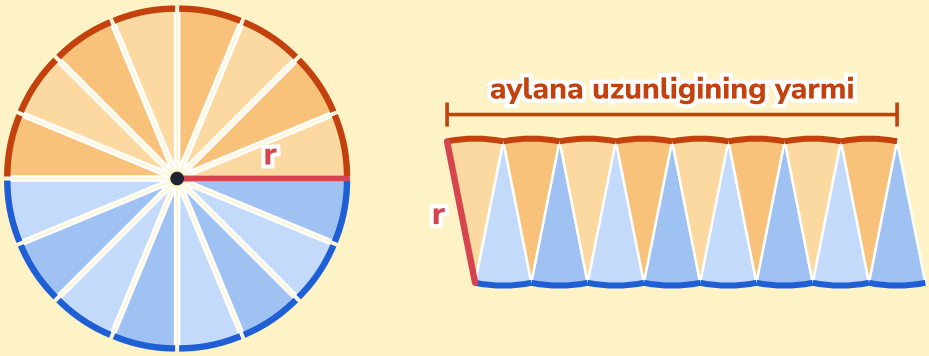
1. Katta kvadrat tomoni r bo'lgan 4 ta kvadratchadan iborat: yuzi $4 \cdot r \cdot r = 4r^2$. Doira undan kichik.
2. Kichik kvadrat 4 ta uchburchakdan iborat. Har biri $r \cdot r$ kvadratchaning yarmi. Kichik kvadratning yuzi $2r^2$. Doira undan katta.
3. Demak, doiraning yuzi $2r^2$ dan katta, $4r^2$ dan kichik — taqriban $3r^2$.



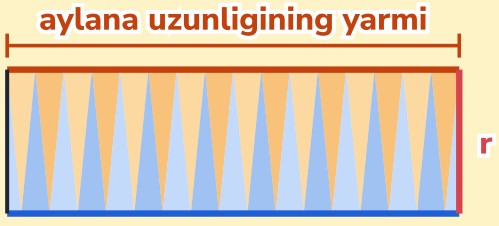
Radiusi 4 cm bo'lsa, $3 \cdot 4 \cdot 4 = 48$. Kataklar bilan esa 52 chiqqan edi. Ikkalasi ham taxminiy-ku!

QOIDA · DOIRANING YUZI

Doirani teng bo'laklarga kesamiz. Bo'laklarni bir qatorga teramiz: birining uchi pastga, keyingisining uchi tepaga qaraydi.



Bo'laklar qancha ko'p bo'lsa, shakl to'g'ri to'rtburchakka shuncha o'xshaydi. Uning eni taqriban aylana uzunligining yarmiga, bo'yi — radiusga teng.



$$\text{Doiraning yuzi} \approx \text{to'g'ri to'rtburchakning yuzi}$$
$$= \text{eni} \times \text{bo'yi}$$
$$= \frac{1}{2} \times \text{aylana uzunligi} \times \text{radius}$$
$$= \frac{1}{2} \times 2 \times \pi \times \text{radius} \times \text{radius}$$
$$= \pi \times \text{radius} \times \text{radius}$$
$$\text{Doira yuzi} = \pi r^2$$

1-MISOL · RADIUS BERILGAN

Doiraning radiusi 3,5 cm.

a) Doiraning yuzini π orqali ifodalang.

b) $\pi = \frac{22}{7}$ deb olib, yuzini hisoblang.

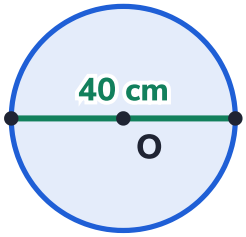


a) $\pi \cdot 3{,}5 \cdot 3{,}5 = 12{,}25\pi$ cm^2.

b) $\frac{22}{7} \cdot 3{,}5 \cdot 3{,}5$. Avval $3{,}5 : 7 = 0{,}5$. So'ng $22 \cdot 0{,}5 \cdot 3{,}5 = 38{,}5$ cm^2.

2-MISOL · DIAMETR BERILGAN

Dumaloq oynaning diametri 40 cm. Oynaning yuzini toping ($\pi = 3{,}14$).



1. Radius: $40 : 2 = 20$ cm.
2. Yuzi: $3{,}14 \cdot 20 \cdot 20 = 3{,}14 \cdot 400 = 1256$ cm^2.

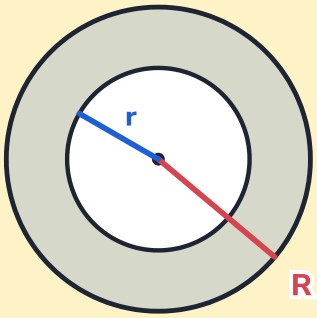
Formulada radius turadi. Diametr berilsa, avval uni 2 ga bo'ling. Yuz kvadrat birliklarda yoziladi: cm^2 yoki m^2.



USUL · HALQANING YUZI

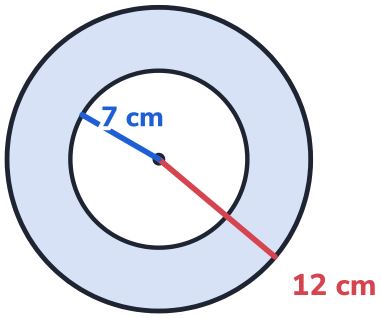
Katta doiradan markazi bir xil bo'lgan kichik doira olib tashlansa, **halqa** qoladi.

1. Katta doiraning yuzini toping.
2. Kichik doiraning yuzini toping.
3. Halqaning yuzi = katta doiraning yuzi − kichik doiraning yuzi.



1-MISOL · IKKALA RADIUS BERILGAN

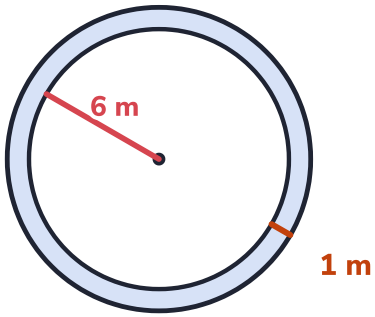
Katta doiraning radiusi 12 cm, kichik doiraniki 7 cm. Bo'yalgan halqaning yuzini toping ($\pi = 3{,}14$).



1. Katta doira: $3{,}14 \cdot 12 \cdot 12 = 452{,}16\ \text{cm}^2$.
2. Kichik doira: $3{,}14 \cdot 7 \cdot 7 = 153{,}86\ \text{cm}^2$.
3. Halqa: $452{,}16 - 153{,}86 = 298{,}3\ \text{cm}^2$.

2-MISOL · HALQANING ENI BERILGAN

Dumaloq hovuzning radiusi 6 m. Uning atrofida eni 1 m bo'lgan yo'lka bor. Yo'lkaning yuzini toping ($\pi = 3{,}14$).



1. Katta doiraning radiusi: $6 + 1 = 7$ m. Yuzi: $3{,}14 \cdot 7 \cdot 7 = 153{,}86\ \text{m}^2$.
2. Hovuz: $3{,}14 \cdot 6 \cdot 6 = 113{,}04\ \text{m}^2$.
3. Yo'lka: $153{,}86 - 113{,}04 = 40{,}82\ \text{m}^2$.

2-QISM

Misollar ishlash

1-MISOL · JAVOB π ORQALI, DIAMETR BERILGAN

Doiraning diametri 2,6 dm. Uning yuzini π orqali ifodalang.

1. Radius: $2{,}6 : 2 = 1{,}3$ dm.
2. $\pi \cdot 1{,}3 \cdot 1{,}3 = 1{,}69\pi$ dm^2. π o'rniga son qo'yilmaydi — javob shu ko'rinishda qoladi.

2-MISOL · $\pi = \frac{22}{7}$, AVVAL QISQARTIRAMIZ

Doiraning radiusi 35 cm. Uning yuzini toping ($\pi = \frac{22}{7}$).

1. $\frac{22}{7} \cdot 35 \cdot 35$. Bitta 35 ni 7 ga bo'lamiz: $35 : 7 = 5$.
2. Qolgani: $22 \cdot 5 \cdot 35 = 110 \cdot 35 = 3850$ cm^2.

Radius 7 ga bo'linsa, $\frac{22}{7}$ qulay: avval qisqartirsak, hisob osonlashadi.

3-MISOL · JAVOBNI YAXLITLASH

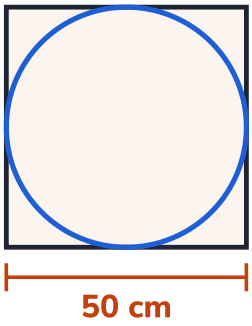
Suv purkagich maysani radiusi 2,3 m bo'lgan doira shaklida sug'oradi. Sug'oriladigan joyning yuzini toping ($\pi = 3{,}14$). Javobni yuzdan birlar xonasigacha yaxlitlang.

1. $2{,}3 \cdot 2{,}3 = 5{,}29$.
2. $3{,}14 \cdot 5{,}29 = 16{,}6106$ m^2.
3. Verguldan keyingi uchinchi raqam 0, u 5 dan kichik: $16{,}6106 \approx 16{,}61$ m^2.

4-MISOL · KVADRATDAN DOIRA KESILDI

Tomoni 50 cm bo'lgan kvadrat shaklidagi taxtadan eng katta doira kesib olindi. Ortib qolgan taxtaning yuzini toping ($\pi = 3{,}14$).

1. Doira kvadratga tegib turadi: diametri kvadrat tomoniga teng, 50 cm. Radius: $50 : 2 = 25$ cm.
2. Doira: $3{,}14 \cdot 25 \cdot 25 = 1962{,}5\ \text{cm}^2$. Kvadrat: $50 \cdot 50 = 2500\ \text{cm}^2$.
3. Ortib qolgan qism: $2500 - 1962{,}5 = 537{,}5\ \text{cm}^2$.





r^2 ni $2r$ deb o'qib yuborish oson. $r^2 = r \cdot r$!

DIQQAT!

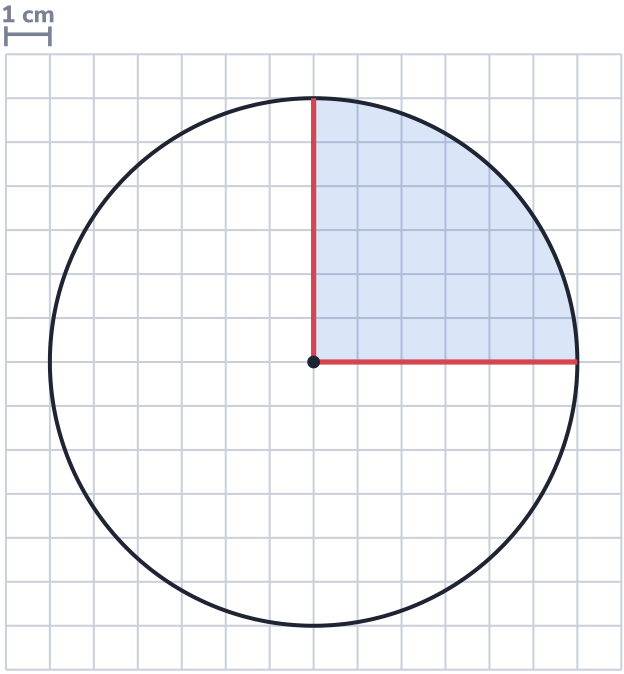
1. $r^2 = r \cdot r$. Masalan, $r = 3$ cm bo'lsa: ✗ $3{,}14 \cdot 3 \cdot 2$ ✓ $3{,}14 \cdot 3 \cdot 3 = 28{,}26\ \text{cm}^2$
2. Diametr berilsa, avval radiusni toping. $d = 12$ cm: ✗ $3{,}14 \cdot 12 \cdot 12$ ✓ $3{,}14 \cdot 6 \cdot 6 = 113{,}04\ \text{cm}^2$
3. Yuz kvadrat birlikda yoziladi: ✗ 113,04 cm ✓ $113{,}04\ \text{cm}^2$
4. Halqada radiuslar ayirilmaydi, yuzlar ayiriladi. 7 m va 6 m: ✗ $3{,}14 \cdot 1 \cdot 1$ ✓ $153{,}86 - 113{,}04 = 40{,}82\ \text{m}^2$
5. $2\pi r$ — aylana uzunligi, πr^2 — doiraning yuzi. Ularni adashtirmang.

3-QISM

Mustahkamlash

1. Bo'sh joyni to'ldiring.
   a) Doira yuzi $= \pi \cdot \ldots \cdot \ldots$
   b) Doira bo'laklari qatorga terilsa, deyarli to'g'ri to'rtburchak hosil bo'ladi. Uning eni ... ga, bo'yi ... ga teng.
   c) Doiraning yuzi $2r^2$ dan ..., $4r^2$ dan ... .

2. Radiusi 6 cm bo'lgan doira katakli qog'ozga chizildi (1 katak — $1\ \text{cm}^2$). Bo'yalgan chorak doiradagi kataklarni sanang: yarmidan ko'pi bo'yalganini butun deb oling. Butun doiraning yuzini chamalang.



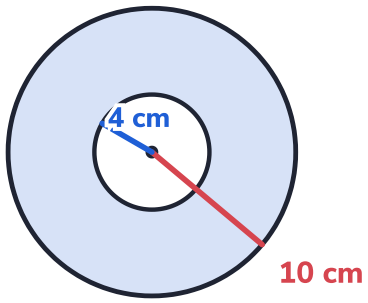
3. Doiraning yuzini π orqali ifodalang.
   a) $r = 3$ cm
   b) $r = 13$ m
   c) $d = 1{,}2$ m
   d) $d = 40$ mm.

4. Doiraning yuzini toping ($\pi = \dfrac{22}{7}$).
   a) $r = 21$ cm
   b) $r = 0{,}7$ m
   c) $d = 2{,}8$ m
   d) $r = 56$ mm.

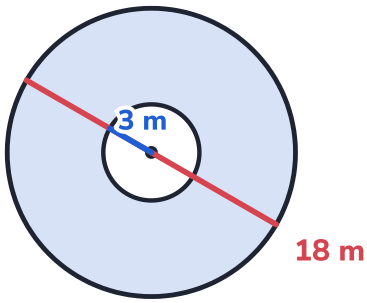
5. Doiraning yuzini toping ($\pi = 3{,}14$).
   a) $r = 3$ m
   b) $r = 0{,}5$ m
   c) $d = 60$ cm
   d) $r = 1{,}2$ dm.

6. Doiraning yuzini toping ($\pi = 3{,}14$).
   a) $r = 1{,}7$ cm — javobni yuzdan birlar xonasigacha yaxlitlang
   b) $d = 9$ m — javobni o'ndan birlar xonasigacha yaxlitlang
   c) $r = 3{,}3$ dm — javobni yuzdan birlar xonasigacha yaxlitlang.

7. Bo'yalgan halqaning yuzini toping ($\pi = 3{,}14$).

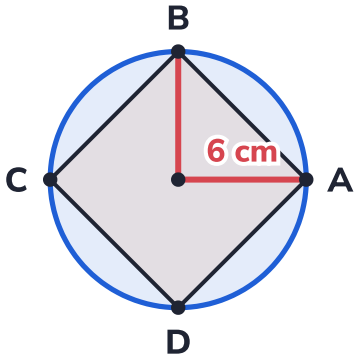


a)



b)

8. Doiraning ichiga kvadrat chizildi: kvadratning uchlari aylanada yotadi. Doiraning radiusi 6 cm. Doiraning kvadratdan tashqaridagi qismi yuzini toping ($\pi = 3{,}14$).



9. Dumaloq gilamning diametri 3 m. 1 m^2 gilam 120 000 so'm turadi. Gilam necha so'm turadi? ($\pi = 3{,}14$)

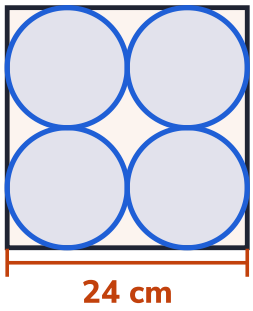
10. Diametri 32 cm bo'lgan bitta katta pitsa yoki diametri 22 cm bo'lgan ikkita pitsa — qaysi birida yeyiladigan joy ko'p? Qanchaga ko'p? ($\pi = 3{,}14$)

11. Doira shaklidagi maysazorning radiusi 2,1 m. Har 1 m^2 ga 40 g urug' sepiladi. Necha gramm urug' kerak? ($\pi = \frac{22}{7}$)

4-QISM

Mustaqil ish

1. Radiusi 4,5 cm bo'lgan doiraning yuzini toping ($\pi = 3{,}14$).
2. Radiusi 4,9 cm bo'lgan doiraning yuzini toping ($\pi = \frac{22}{7}$).
3. Diametri 5 m bo'lgan doiraning yuzini π orqali ifodalang.
4. Radiusi 2,6 m bo'lgan doiraning yuzini toping ($\pi = 3{,}14$). Javobni o'ndan birlar xonasigacha yaxlitlang.
5. Dumaloq stolning radiusi 50 cm. Unga radiusi 70 cm bo'lgan dumaloq dasturxon yopildi. Dasturxonning stoldan osilib turgan qismi yuzini toping ($\pi = 3{,}14$).
6. Tomoni 12 cm bo'lgan kvadrat shaklidagi qog'ozdan eng katta doira qirqib olindi. Qancha qog'oz ortib qoldi? ($\pi = 3{,}14$)
7. Quduqning dumaloq yog'och qopqog'i diametri 1,4 m. Uning ustki tomonini bo'yash kerak. Har 1 m^2 ga 200 g bo'yoq ketadi. Necha gramm bo'yoq kerak? ($\pi = \frac{22}{7}$)
8. **Xatoni toping.** Sardor radiusi 4 dm bo'lgan doiraning yuzini shunday topdi: $3{,}14 \cdot 4 \cdot 2 = 25{,}12\ \text{dm}^2$. Xato nimada? To'g'ri javob qancha?
9. **Qiyinroq.** Tomoni 24 cm bo'lgan kvadrat ichiga 4 ta bir xil doira chizildi (rasm). Doiralar bir-biriga va kvadrat tomonlariga tegib turadi. Kvadratning doiralardan tashqaridagi qismi yuzini toping ($\pi = 3{,}14$).



5-QISM

Javoblar

Mustahkamlash

1-masala

a) $r \cdot r$ (radius · radius); b) eni — aylana uzunligining yarmiga, bo'yi — radiusga; c) katta, kichik.

2-masala

To'liq kataklar — 22 ta, yarmidan ko'pi bo'yalganlari — 6 ta: chorak $\approx 28\ \text{cm}^2$. Doira $\approx 4 \cdot 28 = 112\ \text{cm}^2$. (Formula bilan: $3{,}14 \cdot 6 \cdot 6 = 113{,}04\ \text{cm}^2$.) Ikkita katak deyarli yarmigacha bo'yalgan: ularni ham sanasangiz, $4 \cdot 30 = 120\ \text{cm}^2$ chiqadi. Chamalashda bunday farq bo'lishi mumkin.

3-masala

a) $9\pi\ \text{cm}^2$; b) $169\pi\ \text{m}^2$; c) $r = 0{,}6$ m, $0{,}36\pi\ \text{m}^2$; d) $r = 20$ mm, $400\pi\ \text{mm}^2$.

4-masala

a) $\frac{22}{7} \cdot 21 \cdot 21 = 22 \cdot 3 \cdot 21 = 1386\ \text{cm}^2$; b) $22 \cdot 0{,}1 \cdot 0{,}7 = 1{,}54\ \text{m}^2$; c) $r = 1{,}4$ m, $22 \cdot 0{,}2 \cdot 1{,}4 = 6{,}16\ \text{m}^2$; d) $22 \cdot 8 \cdot 56 = 9856\ \text{mm}^2$.

5-masala

a) $3{,}14 \cdot 9 = 28{,}26\ \text{m}^2$; b) $3{,}14 \cdot 0{,}25 = 0{,}785\ \text{m}^2$; c) $r = 30$ cm, $3{,}14 \cdot 900 = 2826\ \text{cm}^2$; d) $3{,}14 \cdot 1{,}44 = 4{,}5216\ \text{dm}^2$.

6-masala

a) $3{,}14 \cdot 2{,}89 = 9{,}0746 \approx 9{,}07\ \text{cm}^2$; b) $r = 4{,}5$ m, $3{,}14 \cdot 20{,}25 = 63{,}585 \approx 63{,}6\ \text{m}^2$; c) $3{,}14 \cdot 10{,}89 = 34{,}1946 \approx 34{,}19\ \text{dm}^2$.

7-masala

a) $314 - 50{,}24 = 263{,}76\ \text{cm}^2$. b) Katta doiraning radiusi $18 : 2 = 9$ m: $254{,}34 - 28{,}26 = 226{,}08\ \text{m}^2$.

8-masala

Doira: $3{,}14 \cdot 6 \cdot 6 = 113{,}04\ \text{cm}^2$. Kvadrat 4 ta uchburchakdan iborat, har biri $6 \cdot 6 : 2 = 18\ \text{cm}^2$. Kvadratning yuzi $4 \cdot 18 = 72\ \text{cm}^2$. Tashqi qism: $113{,}04 - 72 = 41{,}04\ \text{cm}^2$.

9-masala

$r = 1{,}5$ m. Yuzi: $3{,}14 \cdot 1{,}5 \cdot 1{,}5 = 7{,}065\ \text{m}^2$. Narxi: $7{,}065 \cdot 120\,000 = 847\,800$ so'm.

10-masala

Katta pitsa: $3{,}14 \cdot 16 \cdot 16 = 803{,}84\ \text{cm}^2$. Ikkita kichik: $2 \cdot 3{,}14 \cdot 11 \cdot 11 = 759{,}88\ \text{cm}^2$. Katta pitsada $803{,}84 - 759{,}88 = 43{,}96\ \text{cm}^2$ ko'p.

11-masala

$\frac{22}{7} \cdot 2{,}1 \cdot 2{,}1 = 22 \cdot 0{,}3 \cdot 2{,}1 = 13{,}86\ \text{m}^2$. Urug': $13{,}86 \cdot 40 = 554{,}4$ g.

Mustaqil ish

1-masala

$3{,}14 \cdot 4{,}5 \cdot 4{,}5 = 3{,}14 \cdot 20{,}25 = 63{,}585\ \text{cm}^2$.

2-masala

$\frac{22}{7} \cdot 4{,}9 \cdot 4{,}9 = 22 \cdot 0{,}7 \cdot 4{,}9 = 75{,}46\ \text{cm}^2$.

3-masala

$r = 2{,}5$ m: $\pi \cdot 2{,}5 \cdot 2{,}5 = 6{,}25\pi\ \text{m}^2$.

4-masala

$3{,}14 \cdot 6{,}76 = 21{,}2264 \approx 21{,}2\ \text{m}^2$.

5-masala

Dasturxon: $3{,}14 \cdot 70 \cdot 70 = 15\,386\ \text{cm}^2$. Stol: $3{,}14 \cdot 50 \cdot 50 = 7850\ \text{cm}^2$. Osilgan qism: $15\,386 - 7850 = 7536\ \text{cm}^2$.

6-masala

$r = 6$ cm. Doira: $113{,}04\ \text{cm}^2$. Kvadrat: $144\ \text{cm}^2$. Ortib qoldi: $144 - 113{,}04 = 30{,}96\ \text{cm}^2$.

7-masala

$r = 0{,}7$ m. Yuzi: $\frac{22}{7} \cdot 0{,}7 \cdot 0{,}7 = 1{,}54\ \text{m}^2$. Bo'yoq: $1{,}54 \cdot 200 = 308$ g.

8-masala

Sardor r^2 o'rniga $r \cdot 2$ ni oldi. To'g'risi: $3{,}14 \cdot 4 \cdot 4 = 50{,}24\ \text{dm}^2$.

9-masala

Har bir doiraning diametri $24 : 2 = 12$ cm, radiusi 6 cm. Bitta doira: $113{,}04\ \text{cm}^2$, to'rttasi: $452{,}16\ \text{cm}^2$. Kvadrat: $24 \cdot 24 = 576\ \text{cm}^2$. Tashqi qism: $576 - 452{,}16 = 123{,}84\ \text{cm}^2$.